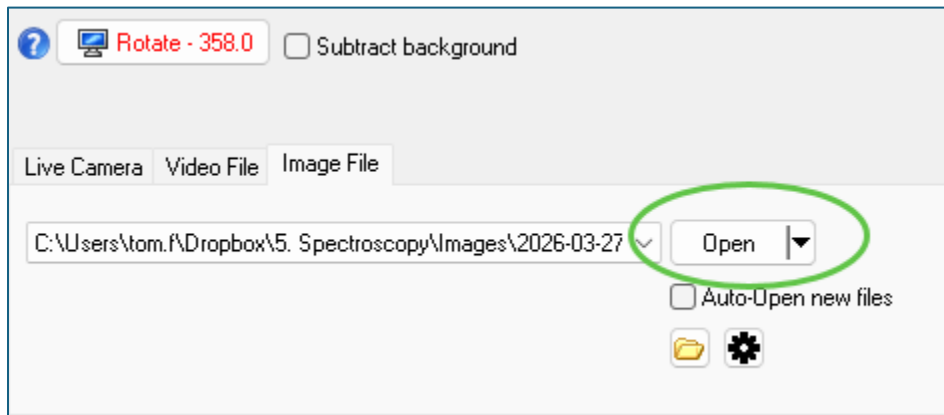


Processing Spectra for BAROSpection

1. Copy your image to a local folder on your computer.
2. Open the image of an A- type star, using the **Open** button in the lower left corner of the screen or by dragging and dropping it into the image window on the left.



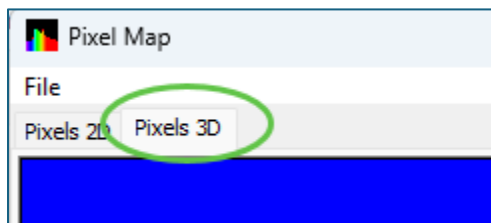
3. If you have trouble seeing the spectrum because the image is a bit dark, open the **Histogram** Stretch by clicking on the



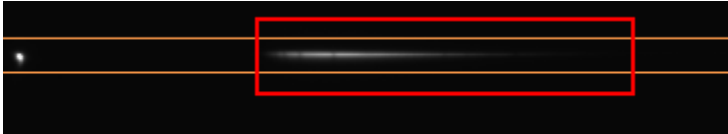
button on the toolbar at the top of the screen.

4. **Checking exposure levels:** Open the **Pixel Map** window with this toolbar button: .

5. Select the **Pixels 3D** tab.

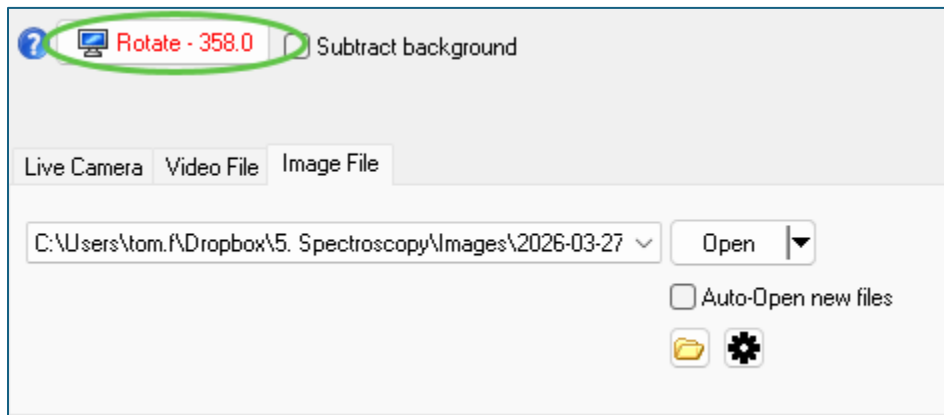


6. Hold down the **Ctrl** key and your left mouse button and draw a box around the spectrum.

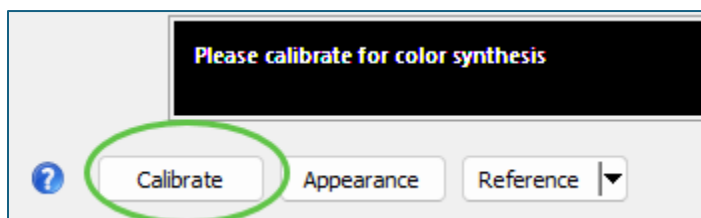


7. Examine the 3D graph to confirm that the graph is not "flat topped," which would indicate the image is over-exposed. The vertical **Flux Intensity** axis's maximum is 65k. Your data should be above 25k, preferably above 40k or 50k.
8. Once completed take a snapshot of the 3D graph window making sure star name is shown and paste to Slack.

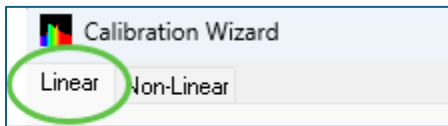
9. Rotate the image if necessary so the spectrum runs horizontally by clicking on the **Rotate** button at the bottom left of the screen.



10. If you want to visually examine the image more closely, you can click on the **Zoom** toolbar button  and then use your mouse roller wheel to zoom, and left click/drag to pan. Double-click resets to the default.
11. Move the orange sampling lines to bracket in your spectrum as close as you can. (You can watch on the Zoom window for a more detailed view.)
12. **Wavelength Calibration:** Click the **Calibrate** that is just below the graphing area



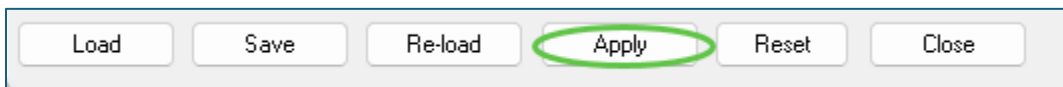
13. Select the **Linear** tab.



14. Enter the pixel value of the zero-order star and its "wavelength" of zero.

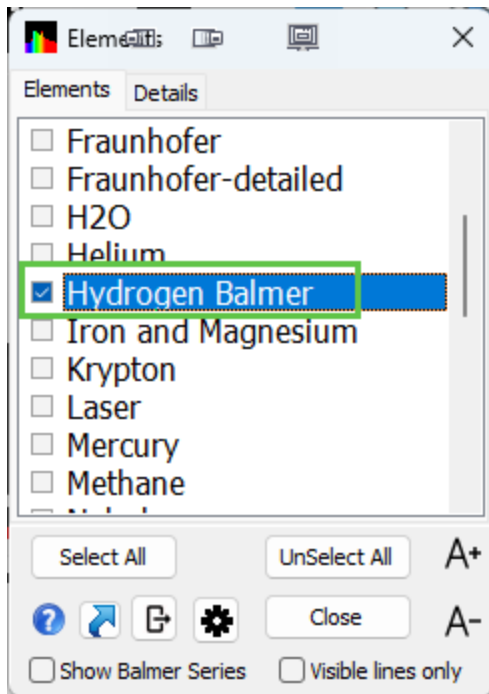
15. For your second point, enter the pixel value of the Hydrogen beta absorption line that is near the peak. And a checkmark in the box for Hydrogen beta.

16. Click the **Apply** button.



17. **Confirm your calibration:** by clicking on the **Elements** toolbar button above the graph 

18. Put a checkmark next to Hydrogen Balmer.



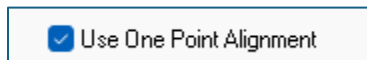
19. Click the floppy icon  to save your profile graph in a text file in the same folder as the original image in step 1.

20. Note down the Angstroms/pixel value that's displayed above the graph. This number is a constant for the current telescope setup.

Angstroms/Pixel: 4.85

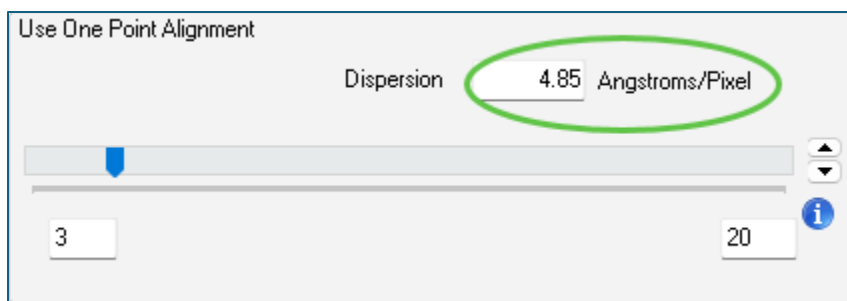
21. Once completed take a snapshot of the profile with element lines making sure star name is shown and paste to Slack.

22. **Wavelength Calibration of all subsequent stars:** Follow the same procedure as above, but when calibrating for wavelength, put a check mark in the **Use One Point Alignment** box on the Calibration Wizard screen.



☒ Use One Point Alignment

23. Make sure that the Angstroms/pixel value is the same as you noted in step 13.

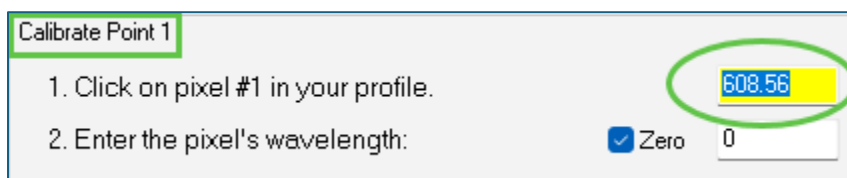


Use One Point Alignment

Dispersion 4.85 Angstroms/Pixel

3 20

24. In the "Calibrate Point 1" section, enter the pixel value in Field 1 for the zero-order star and click **Apply**.



Calibrate Point 1

1. Click on pixel #1 in your profile.

2. Enter the pixel's wavelength:

☒ Zero 0

25. Confirm proper calibration by using the **Elements** window as you did previously, selecting the expected star type.
26. Once completed take a snapshot of the profile with element lines making sure star name is shown and paste to Slack.

